

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Executed installation, maintenance, and repair of electromechanical semiconductor equipment at key customer sites, achieving a **98% on-time completion rate** for scheduled preventative maintenance tasks.
- Resolved complex technical issues on advanced semiconductor manufacturing equipment through disciplined troubleshooting and precise problem-solving, reducing average tool downtime by **15%**.
- Delivered structured customer training sessions on equipment operation and maintenance, resulting in a **20% decrease** in repeat customer-reported issues related to user error.
- Collaborated with global engineering teams to escalate and resolve critical issues, contributing insights that improved the overall reliability of fielded equipment by **10%**.
- Managed and executed change-control processes for system configuration updates and software deployments on enterprise business systems, ensuring **100% adherence to change windows** and minimizing operational disruption.
- Developed Python scripts to automate data extraction and reporting for operational dashboards, reducing manual reporting time by **40 hours per month** and improving data accuracy.

SKILLS

Field Service Engineering – Installation, Decommissioning, Preventative Maintenance, Repair, Troubleshooting, Customer Training, Technical Support, Site-Based Operations

Semiconductor Equipment – Electromechanical Systems, Vacuum Control, Gas Systems, RF Plasma Systems, Electropneumatic Controls, Microprocessor Controllers, Robotics

Software & Tools – Python, Java, Spring Boot, SQL (PostgreSQL, MySQL, Oracle), AWS, Docker, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, R, pandas, NumPy, scikit-learn

Core Engineering – Electrical Schematics, Mechanical Troubleshooting, Electrical Troubleshooting, Linux, Shell Scripting, REST APIs, Change Management, UAT Coordination

Compliance & Safety – Safety Protocols, Cleanroom Standards, Regulatory Awareness, GDPR Compliance, Audit Trails

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Advanced Semiconductor Equipment Remote Diagnostics Platform

- Engineered a **remote diagnostics platform** using Python and Flask, enabling real-time monitoring and troubleshooting of deployed semiconductor equipment, which reduced on-site service visits by **25%**.
- Integrated **AWS IoT Core** and **MQTT** for secure, high-volume telemetry data streaming from distributed equipment fleets, processing over 1 million data points daily with less than 50ms latency.
- Developed interactive dashboards in **Power BI** visualizing equipment health metrics, predictive maintenance indicators, and performance trends, empowering field engineers with actionable insights for proactive issue resolution.
- Implemented automated alert generation based on anomaly detection algorithms (scikit-learn) for critical system parameters, preventing potential failures and improving tool uptime by an average of **18%** across monitored assets.
- Managed CI/CD pipelines using **Jenkins** and **Docker** for continuous deployment of platform updates, ensuring rapid iteration and stability for the core diagnostic services.

Financial Services Transaction Monitoring Dashboard

- Supported a **payments operations dashboard** for a financial-services client, focusing on transaction-flow monitoring and operational alerting for failed payment messages.
- Maintained production environments with strict SLAs, ensuring robust **audit trails** and adherence to regulatory requirements through diligent monitoring and incident response.
- Automated data validation and reconciliation workflows using Python and SQL against PostgreSQL and Oracle databases, enhancing the accuracy of transaction status reporting.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)